

Augmenting Everyday Objects into Personal Robotic Devices

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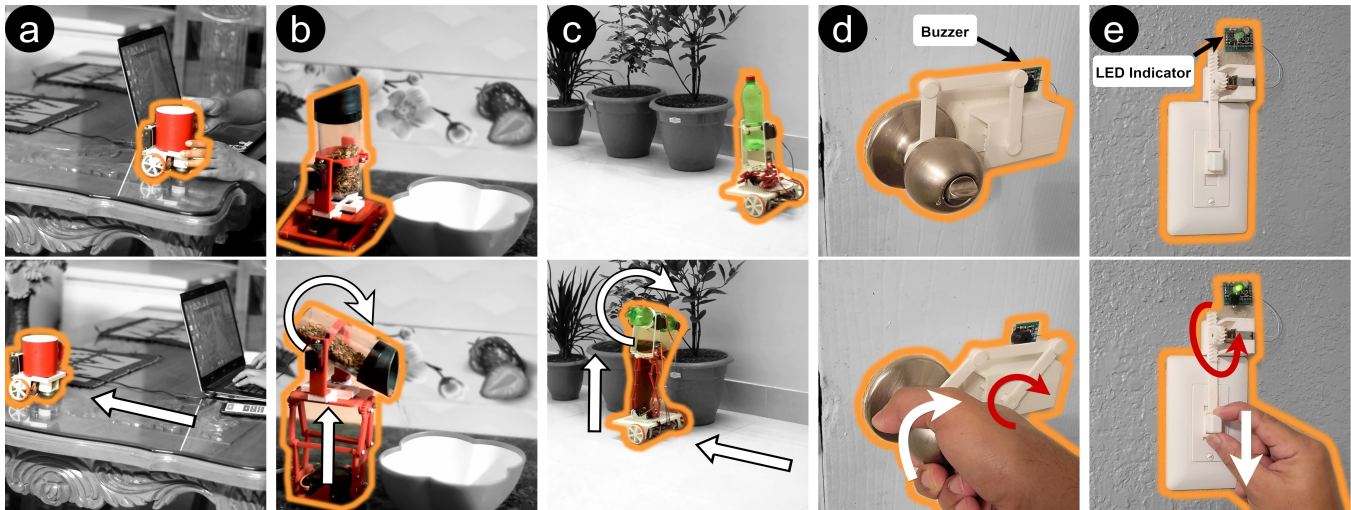


Figure 1: Our daily essential objects have a great potential to become smart robotic devices, from domestic assistive robots (a-c) to self-sustainable smart objects (d-e), leveraging personal fabrication and end-user programming of robotics.

ABSTRACT

Augmenting familiar physical objects has presented great potential in upgrading their functions by automation, granting aesthetics, and even changing access. The recent celebration of success in personal fabrication has brought novices where they can augment everyday objects, from automating routine tasks with mobilized smart devices to devising self-sustaining smart objects by harvesting energy from involved daily interactions, for example. While the overall process involves a line of steps of capturing specifications, design mechanisms and fabricating the parts, it remains challenging for non-experts as it demands domain expertise in robotics, design, programming, and even mechanical engineering.

We introduce a series of *augmented robots*, smart domestic devices that are augmented from everyday objects, leveraging personal fabrication to assist daily essential interactions. Through user-demonstration of desired motions, 3D printed attachment mechanisms are auto-generated to build personal robotic devices that automate routine tasks and harvest energy.

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1 INTRODUCTION

Digital fabrication has the potential to reconfigure our everyday physical world for diverse purposes by augmenting them into robotic devices; to change access [Chen et al. 2016; Kim et al. 2017], to automate otherwise passive home environment [Arabi et al. 2022; Li et al. 2019], to harvest energy from daily activities [Yang et al. 2022] that are great sources to power smart devices yet often overlooked. While the recent success of low-cost fabrication technology and increased end-user capabilities to build and program personal robotic devices promised a future everyone can take the pleasure of DIY robotics at hand, varying specifications and custom need to construct augmented robotics from daily objects present a huge knowledge gap to non-experts, requiring domain expertise to capture, specify, then decipher the requirements to uncover solutions.

We introduce a series of domestic augmented robotic devices that are custom-fabricated from the user's demonstration by capturing all design requirements. Using currently well-adopted smartphones and wearables (e.g., smartwatch) that nowadays come with native IMU sensors, non-experts can record desired motion by demonstration. Using machine learning techniques (e.g., LSTM), sensor values logged into the device are classified and segmented to reconstruct trajectories to readily auto-generate 3D printable mechanisms. We

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also present an end-user toolkit, enabling novice users to utilize the full expressive design capability to augment familiar everyday objects into domestic smart objects with unique functions. From a one-shot capturing of user demonstration, the tool identifies the target object's geometry, motion paths, and other critical information needed for generating 3D print-ready files, which will further provide authoring features to enable fine-tuning.

2 AUGMENTED ROBOTIC DEVICES

2.1 Self-organizing coffee mug

Figure 1a illustrates a self-organizing coffee mug. The user takes a photo of the coffee mug and records the desired motion with the Mobiot app running on the smartphone. Fetching a 3D model of the coffee mug from a prebuilt repository that provides geometry information, the toolkit parses the trajectory information and the type of actions from the IMU data captured by the app. The toolkit editor then offers the user to optionally fine-tune, and once happy with the editing, it generates the 3D printable mechanisms, creates a list of electronics to be attached, and auto-generates the Arduino code to mobilize the mechanism. Now, the hot coffee mug self-organizes itself to get rid of any accidental spill.

2.2 Self-serving spice jar

Dreaming of a self-serving robot kitchen, regular, passive spice jars on the table can be transformed into self-serving jars for a dinner party. The jar could be either scanned from the demonstration to fetch the corresponding 3D model or uploaded directly to design augmentation mechanisms. A user can demonstrate the motion with a spice jar on the smartphone that has native sensors running to log a series of motions. After assembling the rotation and the scissor lift mechanisms with the target, it is now ready to welcome guests as depicted in Figure 1b.

2.3 Auto-planting water bottle

Figure 1c illustrates a self-planting water bottle, for a user who plans for a long international trip to attend a friend's wedding. The user might want to design an auto-watering system to keep her plants alive. An otherwise passive water bottle may need to start its weekly journey from a faucet in the kitchen to fill water on a regular basis, relocate to rooms and corners where the user decorated the indoors with plants, adjust its height and tilt near each pot to pour water. From a user's recording of these needed motions by simply iterating through her daily routine of watering, the complex motion paths are recorded, parsed, and slightly adjusted upon the user's re-programming. The leftover to the user is the downloading of all auto-generated parts that constitute three mechanisms for roaming, lifting, and tilting. As needed code to operate these motions is auto-generated subsequently, the user can assemble the mechanisms to augment the bottle and upload the code. While away, the user is also able to trigger watering using a smartphone ad-hoc.

2.4 Battery-less door alert

Figure 1d illustrates a mechanism to harvest energy from twisting a doorknob. A user could be initially unaware of its motion as a potential energy source. Once the user runs the E3D app on the

smartwatch early in the morning, it starts to log activity data. At the end of a day-long recording, the activity data is sent to the E3D toolkit and parsed to obtain salient motion features. The user can earn a list of activities that are capable of energy harvesting, and become knowledgeable about twisting as a good source of energy. As the toolkit automatically estimates the range of motion from the trajectory reconstruction, no additional information is needed from a user. After 3D printing, the user attaches a buzzer to make a door alert that beeps when the doorknob is twisted to open.

2.5 Self-powered switches with visual capability

While a short range of motion, such as activating light switches, a door latch locker, a hand soap pump, etc., can provide a small amount of energy per activity, these actions can accumulate substantive energy through multiple, preferably more frequent activation. Figure 1e presents an augmented harvesting mechanism that can yield energy from turning on and off the light for multiple day-time actuation. A simple toggling of the light switch collects the energy, saves it into a supercapacitor attached to it or to a battery, then finally, feeds a low-power LED to assist the users with night low-vision and children in locating the switch.

3 END-USER TOOL

The toolkit takes two inputs from the physical world for augmentation. (i) From the captured input image it runs object detection to fetch a 3D model which provides the geometry information, and (ii) the smartphone app records the demonstration using the native IMU sensor to estimate the trajectory that is typed into translation, rotation, and lifting motion, based on which, the toolkit uses our parametric design library to generate the mechanisms. Likewise, it delivers motion planning data to auto-generate the code to mobilize the object with desired motion. A GUI offers the user to make optional adjustments. Afterward, it lists required electronics and provides assembly instructions. The backend, frontend, and parametric designs are created using Python, Javascript, and OpenScad.

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